Plant methods, reference assemblies and annotation releases

The completed Arabidopsis comparisons do not establish a better replacement for the current BWA-MEM2/MACS3 defaults. BWA closely reproduces the baseline; Bowtie2 changes PE DAP peaks substantially in this configuration, and higher read retention does not consistently improve independent evidence. Reference and annotation choices also affect the measurements and must be reported explicitly.

This is an interim report. The four-species panel is running; the table below separates completed measurements from pending comparisons. No pending experiment is counted as evidence for a package choice.

Coverage

Data	Depth caps per library	Methods / references	Evidence available
A. thaliana SE DAP: FUS3, ABI5, assigned control	1M and 5M templates	Four methods on TAIR10	16 completed method/repetition cases; independent PBM motifs at both depths
A. thaliana PE DAP: four treatments, assigned control	1M and 5M pairs	Four methods on TAIR10	16 completed cases; these protocol-optimization libraries are not biological replicates
A. thaliana bulk ATAC: two biological replicates	1M and 5M pairs	Eight methods on TAIR10	32 completed cases; common metrics, replicate overlap and held-out FRiP
A. thaliana reference comparison	1M and 5M pairs	TAIR10 / Col-CEN v1.2	Completed ATAC baselines at both depths; reciprocal-coordinate comparison at 1M; IDR at 1M and Col-CEN 5M
A. thaliana annotation comparison	1M pairs	TAIR10 / Araport11 annotations on Col-CEN	Completed using the same aligned reads
A. thaliana additional reference/depth cases	1M and 5M	DAP SE/PE and ATAC reference baselines	Running; only completed result manifests enter the report
A. lyrata DAP	1M and 5M caps	Four selected TF families; legacy and study reference distributions	References acquired and fully scanned; read preparation and method runs pending

Sources and reproducible commands

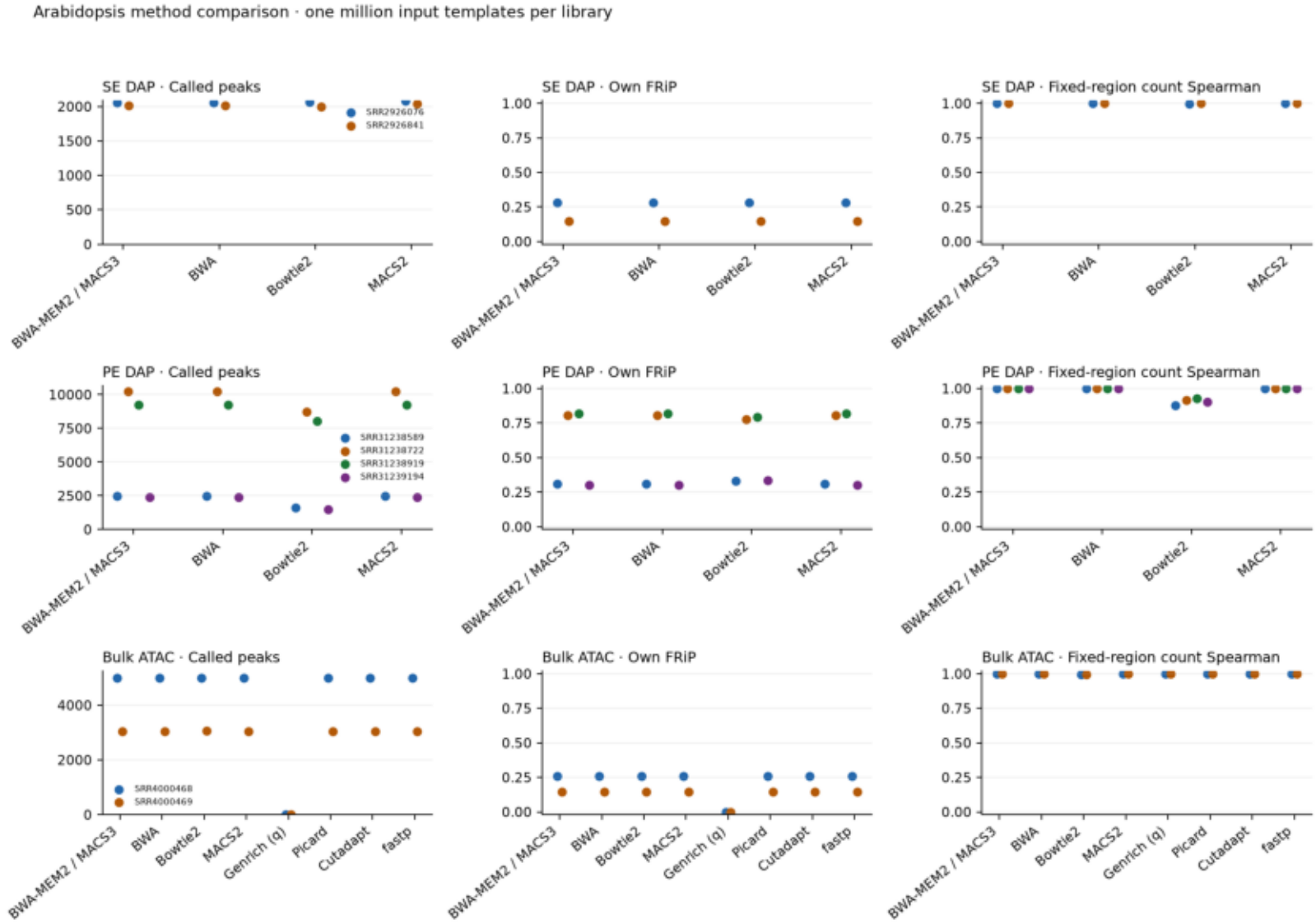
Data	Depth caps per library	Methods / references	Evidence available
Sorghum DAP	1M and 5M pairs	Four selected TF families; Sorbi1 / BTx623 v5 / RTx430 v2	Complete reference sequences acquired; read preparation and method runs pending
Maize bulk ATAC	1M and 5M pairs	Two biological replicates; AGPv4 / NAM5	References acquired; full-library preparation and method runs pending

The 64 completed method cases comprise three technical repetitions at 1M and one at 5M. Technical repetitions measure execution variation; they are not extra biological replicates. Low-yield libraries retain their actual count. In particular, the new A. lyrata EIL library contains 524,758 paired templates, so its 1M and 5M caps will contain the same complete library.

The original comparison report contains the synthetic known-answer tests and complete Genesis/nf-core ATAC comparison. The four-species recipe records data selection, source formats, exact references and reproduction commands.

Method quality and cost

Per-library peak counts, FRiP and fixed-region correlations



Each point is one treatment or biological ATAC replicate; controls are not plotted. Quality was identical across three technical repeats.
FRiP uses SE read spans or PE outer-fragment spans / retained spans; peak widths and coverage are in quality-summary.tsv.
Missing correlation is omitted, never replaced by zero. Baseline regions are a fixed comparison set, not truth. Genrich uses its recorded q-value operating point.

Sources and reproducible commands

Complete quality table includes metric definitions, versions, units, denominators, peak union bases and median widths. Own FRiP uses retained SE read spans or PE outer-fragment spans. Comparisons on baseline regions measure agreement, not biological truth.

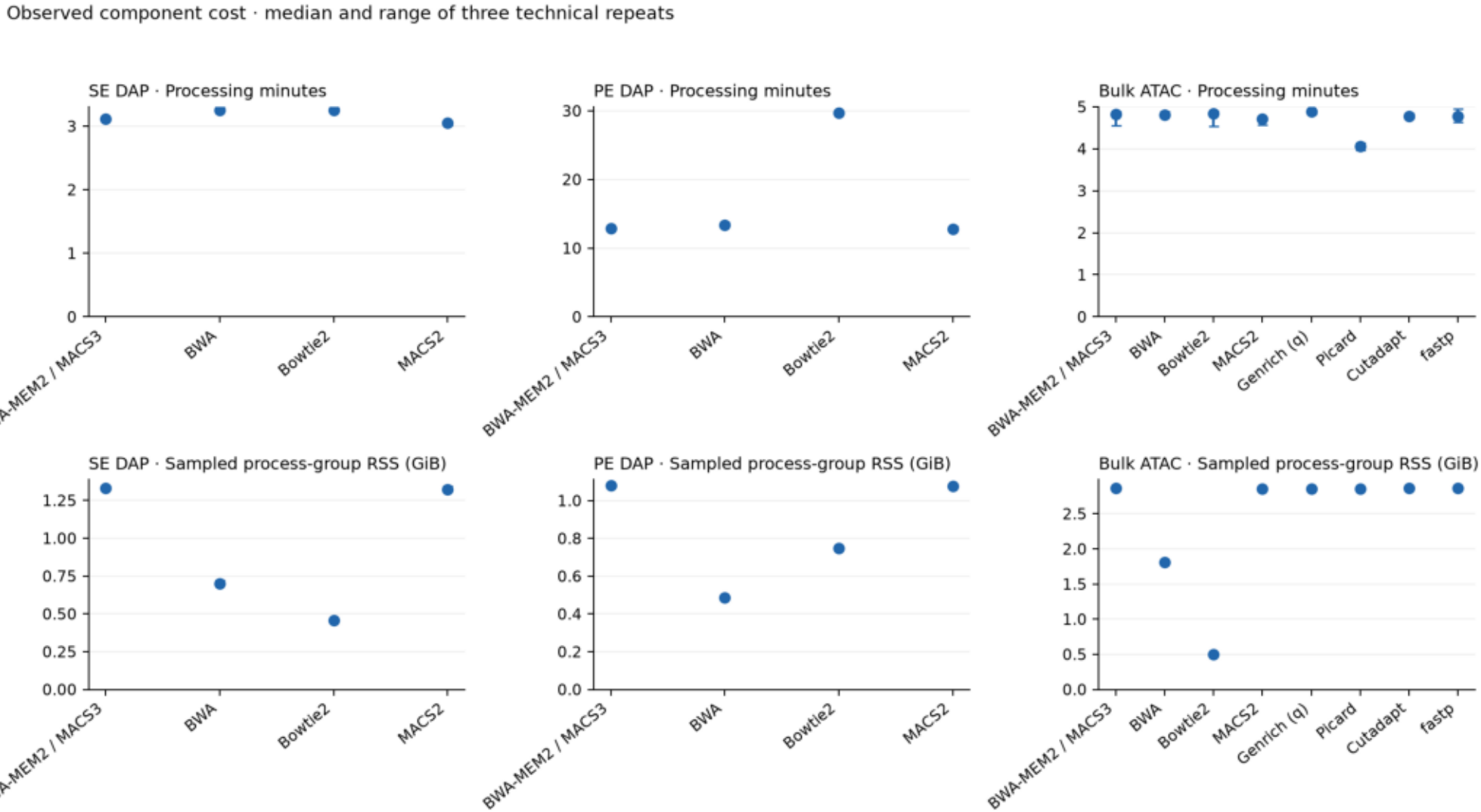
At 1M PE DAP pairs, Bowtie2 changes the low-yield treatments more than the high-yield treatments. This is an observed difference, not a demonstration of incorrect peaks:

Library	Baseline peaks	Bowtie2 peaks	Peak base-pair Jaccard	Fixed-region count Spearman
SRR31238589	2,470	1,595	0.5785	0.8793
SRR31239194	2,398	1,475	0.5611	0.9012
SRR31238722	10,222	8,734	0.8290	0.9152
SRR31238919	9,227	8,046	0.8493	0.9269

Independent PBM motif support remains similar across the tested SE DAP methods at the deeper cap. For FUS3, 466/998 baseline top-ranked matched windows contain the motif, compared with 465/998 for Bowtie2 and 468/998 for MACS2. For ABI5 the counts are 581/1,000, 581/1,000 and 582/1,000, respectively. The FUS3 set is explicitly incomplete: two selected windows could not obtain a matched control. These window counts are not independent TF experiments. Background hits, all eligible windows and exclusions are retained in the motif table.

Sources and reproducible commands

Observed time and memory for each component comparison



All libraries in each cohort; indexing and common evaluation excluded. Index costs are reported separately.
Fourteen tool CPUs, 80 GiB per experiment; randomized method order; shared filesystem and OS caches. Error bars are observed ranges, not confidence intervals.
RSS is a sampled sum for the tool process group, not whole-host peak memory. These are component costs, not complete workflow timings.

Sources and reproducible commands

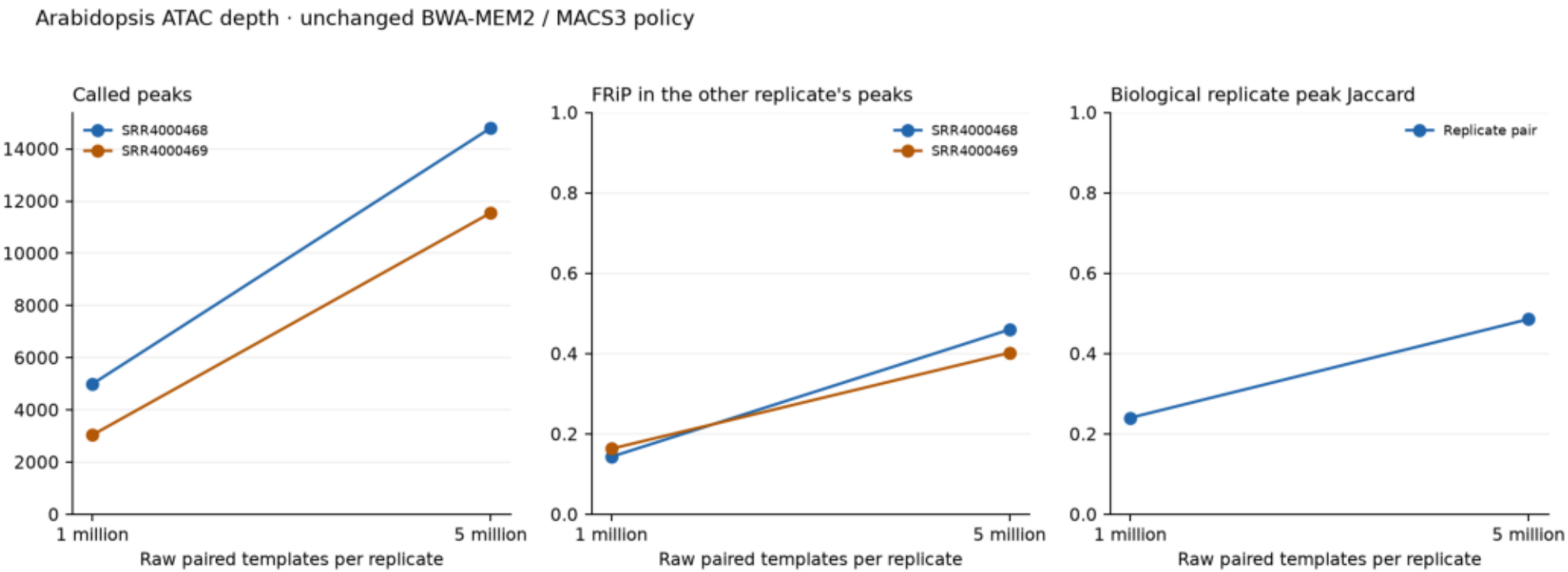
Time is the median and observed range of three fresh technical repetitions on the same inputs. It includes processing all libraries in a cohort but excludes indexing and common evaluation. It also excludes full workflow QC and coverage tasks. Memory is sampled process-group RSS, not whole-host peak RAM.

Cohort, 1M cap	Baseline processing seconds, median [range]	Selected alternative
SE DAP	187.0 [185.5–187.7]	Bowtie2: 194.8 [194.4–197.3]
PE DAP	772.1 [766.1–772.8]	Bowtie2: 1,782.2 [1,778.9–1,785.2]
Bulk ATAC	289.3 [273.2–293.7]	Picard: 243.2 [238.1–248.2]

Picard is a performance candidate for further testing: it gives nearly identical scientific observations here and shorter observed processing time. Its behavior and resource use on the larger maize genome are still pending. A single small plant cohort does not establish the best duplicate marker for all datasets. The cost observations retain every repetition; index measurements remain separate.

Depth and biological reproducibility

ATAC depth, peak discovery and held-out reproducibility



Nested, seeded subsets of two biological replicates on TAIR10. The five-million depth was measured once per method.
Held-out FRiP denominator is retained fragments in the evaluated replicate; Jaccard uses peak base-pair intersection / union.
Increased depth changes both peak discovery and covered bases; these observations do not establish a universal plant QC threshold.

Sources and reproducible commands

TAIR10 baseline	SRR4000468, 1M	SRR4000468, 5M	SRR4000469, 1M	SRR4000469, 5M
Retained fragments / input pairs	0.458813	0.452141	0.565032	0.553789
Peaks	4,997	14,805	3,043	11,551
FRiP in the other replicate's peaks	0.143339	0.460099	0.163293	0.402303
TSS center / flank ratio	3.6741	3.6863	2.7609	2.7805

Replicate peak base-pair Jaccard rises from 0.2404 at 1M to 0.4858 at 5M. Peak coverage also increases, so higher held-out FRiP must be read together with that coverage. TSS enrichment is comparatively stable. This supports evaluating depth and reproducibility together rather than ranking tools by own FRiP alone.

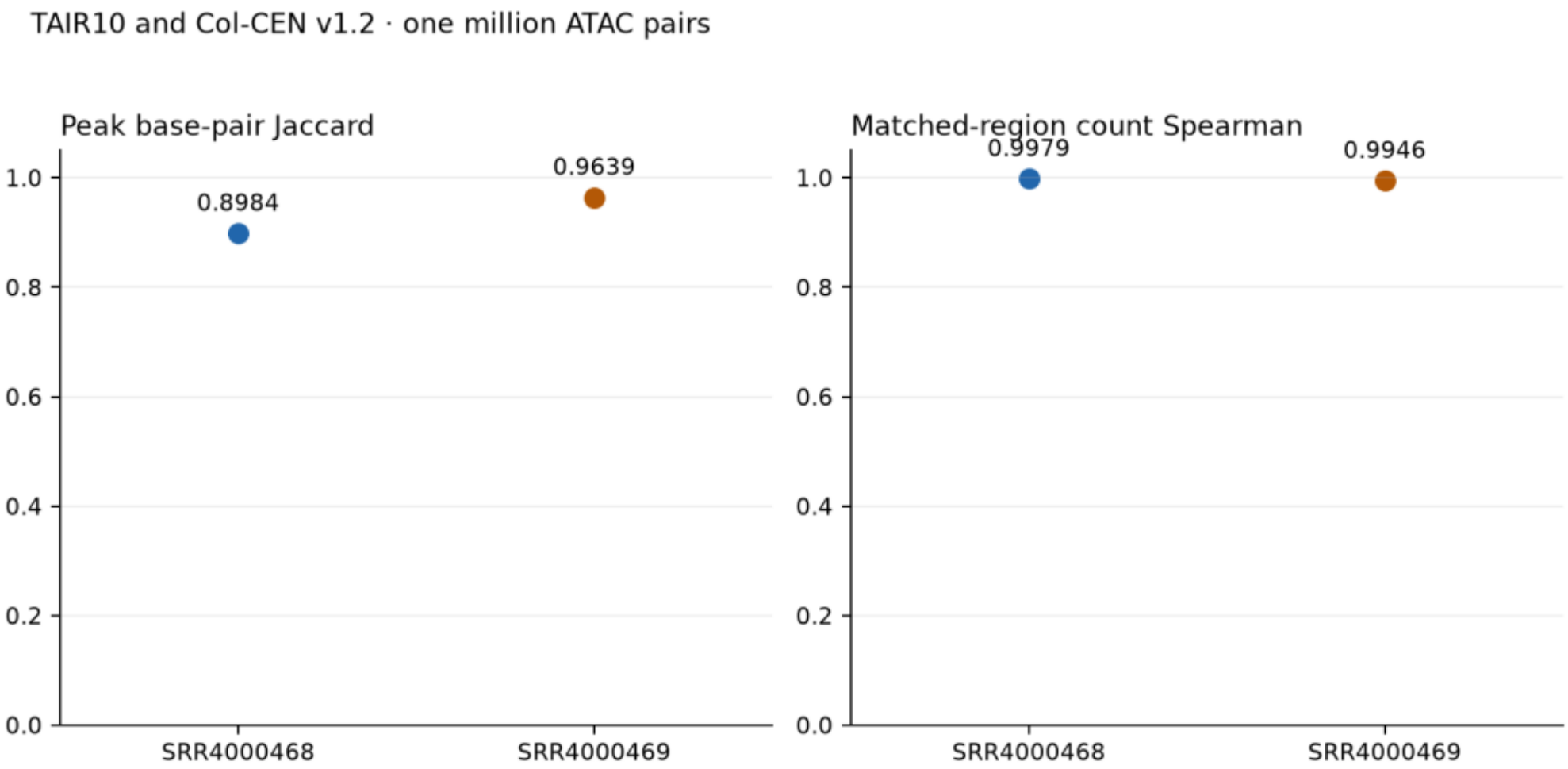
Library complexity is measured before duplicate exclusion using aligned outer fragment coordinates. The baseline NRF values at 1M are 0.99639 and 0.99520; at 5M they are 0.98273 and 0.97605. NRF is distinct fragment-coordinate groups divided by eligible fragments. PBC1 and PBC2, the eligible-fragment denominator, duplicate counts, organellar fractions and fragment distributions are retained in each library-qc.tsv. This coordinate definition is not identical to every ENCODE implementation based on unclipped read starts. No human-cell threshold is applied as a plant acceptance rule.

Reference assembly comparison

The first completed assembly comparison uses the exact same 1M ATAC read subsets, both-mate MAPQ30 selection, duplicate/organellar exclusion and MACS3 policy. Genome-size values are derived separately from each declared reference, so this comparison includes the reference's effect on the peak-calling background model.

Library	Reference	Retained / input pairs	Peaks	Own fragment FRiP
SRR4000468	TAIR10	0.458813	4,997	0.258672
SRR4000468	Col-CEN v1.2	0.459326	5,596	0.273912
SRR4000469	TAIR10	0.565032	3,043	0.146647
SRR4000469	Col-CEN v1.2	0.565606	2,977	0.143561

Concordance after reciprocal mapping between TAIR10 and Col-CEN



Same reads and policy; unique reciprocal assembly mappings only. At least 95% aligned bases in both directions and 95% round-trip span overlap. See the mapping table for all exclusions. Concordance is not biological accuracy.

Sources and reproducible commands

Peaks are compared only after bidirectional assembly alignment and reciprocal coordinate conversion. At least 95% of each interval must align in each direction, with a unique placement and at least 95% reciprocal span overlap. Multiple and failed mappings are excluded and counted, not silently assigned to the first hit.

Library	TAIR10 peaks accepted / total	Col-CEN peaks accepted / total	Projected peak Jaccard	Matched-region fragment Spearman
SRR4000468	4,992 / 4,997	5,586 / 5,596	0.8984	0.9979
SRR4000469	3,040 / 3,043	2,968 / 2,977	0.9639	0.9946

Every mapping outcome and paired count is available. These measurements show strong agreement on eligible regions; they do not establish which assembly is biologically more accurate. The supplied minimap2 alignments bound the uniqueness claim. Repeats and unrepresented secondary alignments remain limitations.

The completed deeper baselines show similar within-reference replicate agreement: peak base-pair Jaccard is 0.4858 on TAIR10 and 0.4878 on Col-CEN. The following values use 5,000,000 input pairs per library. Reciprocal-coordinate comparisons at this depth remain pending.

Library	Reference	Retained fragments / input pairs	Peaks	Own fragment FRiP
SRR4000468	TAIR10	2,260,707 / 5,000,000	14,805	0.528317
SRR4000468	Col-CEN v1.2	2,263,583 / 5,000,000	15,343	0.535342
SRR4000469	TAIR10	2,768,943 / 5,000,000	11,551	0.413543
SRR4000469	Col-CEN v1.2	2,771,657 / 5,000,000	11,930	0.418713

The Col-CEN 5M measurements include peak coverage, held-out FRiP and metric denominators. More peaks and a higher own FRiP alone do not establish an assembly's accuracy.

Sources and reproducible commands

The A. lyrata audit also prevents a misleading comparison: all 695 contigs, totalling 206,667,935 bases, have the same case-insensitive sequence multiset in the legacy Ensembl v1.0 and study Phytozome v2.1 distributions. The large chromosomes are renamed between distributions. This is not evidence of a newer genome sequence; annotation, naming and ordering are separate variables. All 695 contigs have a unique full-sequence match, so this comparison uses exact contig identity mappings rather than an estimated assembly alignment.

For maize, both official MaizeGDB chain files passed publisher MD5 checks and validation against the selected AGPv4/NAM5 FASTA contig lengths and alignment blocks. The forward and reverse files contain 1,003 and 2,413 chains respectively. NAM5 chromosome aliases are explicit. This validates the mapping inputs; maize peak concordance still requires the pending sequencing runs.

Annotation sensitivity

These scores use the same Col-CEN BAMs. Only the supplied annotation changes:

Annotation / TSS set	Eligible unique TSS positions	SRR4000468 ratio	SRR4000469 ratio
Araport11, all transcript starts	46,558	3.6894	2.7517
TAIR10, all transcript starts	32,986	3.8027	3.2096
Araport11, common single-start genes	18,428	3.3807	2.4075
TAIR10, common single-start genes	18,428	3.1585	2.7602

There are 18,430 common genes with one unique TSS in each annotation, on the same contig and strand; two edge positions lack full ±2 kb windows. Their TSS positions may differ between releases. Coding and noncoding transcript types are explicit. The all-transcript sets contain different genes and isoforms, so they are not matched denominators. A higher ratio cannot by itself establish a better annotation.

The score is the single center bin divided by the mean of the outer 100 bins on each side of the strand-aware ±2 kb Tn5 cut profile. The full profiles and gene selection are retained. Annotation release, extraction rules, eligible positions and the precise enrichment definition must accompany any quoted TSS score.

Genrich operating points and exploratory IDR

Genrich's q=0.01 comparison produced valid zero-peak outputs at both Arabidopsis depths. The logs report all q-values equal to one. Its documentation describes this condition when no p-value is below the reciprocal genome length. This result does not show that Genrich is broken or unsuitable for every plant dataset.

A separate diagnostic uses its documented native p=0.01 operating point on the same Col-CEN 1M BAMs. It produces 6,095 and 2,871 peaks, with own FRiP 0.333293 and 0.171750. Their base-pair Jaccards against the MACS3 baseline are 0.4062 and 0.3245. These p-values are not equivalent to the original q-values, and higher FRiP does not establish higher accuracy. The diagnostic has not changed production policy. See the caller evidence.

Pairwise IDR is evaluated separately using permissive MACS3 p=0.01 calls, ranked by signal value, with IDR 2.0.4.2 and seed 60141:

Assembly and depth	Input peaks, replicate 1 / 2	Merged output regions	Global IDR ≤0.05 regions
TAIR10, 1M	11,385 / 8,985	6,158	4,205
Col-CEN v1.2, 1M	13,610 / 10,588	7,357	2,325
Col-CEN v1.2, 5M	19,046 / 15,857	13,545	10,530

The deeper Col-CEN run increases the number of reproducible ranked regions. Its complete diagnostic record also retains the separate Genrich operating points: q=0.01 yields zero peaks, while native p=0.01 yields 5,292 and 2,168. TAIR10's matched 5M IDR assessment remains pending.

The IDR model fits and input rank distributions differ, even where the stricter baseline peak sets agree strongly. These are exploratory depth-limited pairwise results, not an assembly ranking or an ENCODE-style acceptance decision. Pooled and pseudoreplicate rescue/self-consistency ratios have not been computed.

Integrity, validation and remaining work

Original archives for all 64 method cases were checked member by member against their stored SHA256 and byte sizes. New public archive derivatives replace local path prefixes and record separate original/published hashes. Original reads, BAMs, private configuration and credentials are not published. The native ATAC baseline also matches the previously executed production workflow for the recorded scientific observations; that does not imply identical BAM bytes or cost.

The expanded reference tests cover invalid manifests, changed FASTA hashes, renamed/reformatted contigs, coordinate bounds, strand handling, annotation selection and stale bundles. Independent UCSC liftOver fixtures check both strands, insertions/deletions, partial intervals and multiple placements. Streaming sampling tests check nested selections, exact replay, truncated data and byte agreement with ordinary FASTQ sampling. The full project regression suite passed using the installed pinned Nextflow plugin cache after the online plugin repository was unavailable. No scientific expectation was relaxed.

The larger DAP and maize matrices, 5M reference concordance, broader independent motif evidence and full-depth biological validation remain incomplete. The maize comparison omits its publication's gDNA controls; multiDAP comparisons retain the repository's assigned controls rather than reproducing the paper's merged plate controls. Synthetic fixtures validate software, not real biological accuracy. No universal plant QC cutoff or aggregate quality score is assigned.

Download the report PDF, the figure-only PDF or the machine-readable evidence to inspect definitions and denominators before selecting a different package or reference for a study.